LECTURE NOTES

Lecture 1.1: Control Systems and Feedback

Week 1 – DC Motor Speed Control

Class Purpose

This lecture introduces the basic structure of a control system and explains why feedback is useful. The DC motor speed-control system will be used as the common example throughout the course.

Main question for today: How can we make a DC motor stay near a desired speed when the load or operating condition changes?

75-Minute Class Plan

Time	Topic or activity
0–10 min	Motivation and examples of control systems
10–20 min	Components and notation of a control system
20–32 min	Open-loop control and its limitations
32–48 min	Closed-loop feedback and error calculation
48–58 min	Disturbance rejection
58–67 min	Basic performance vocabulary
67–75 min	In-class engineering problem and summary

Learning Objectives

By the end of this lecture, students should be able to:

1. explain the purpose of a control system;
2. identify the reference, controller, plant, input, output, and feedback signal;
3. distinguish between open-loop and closed-loop control;
4. calculate the feedback error for a motor-speed example;
5. explain why feedback helps reject disturbances;
6. use basic performance vocabulary such as transient response, steady-state value, and steady-state error.

1. Motivation: Why Control?

A control system makes a physical system behave in a desired way. Controls appear in many engineering systems.

System	Desired quantity	Actuator	Measurement
Cruise control	Vehicle speed	Throttle	Speed sensor
Thermostat	Temperature	Heater or AC	Temperature sensor
Drone altitude	Height	Motors	IMU or altitude sensor
Robot wheel	Wheel speed	DC motor	Encoder

In this course, the main physical system is a DC motor. The goal is to make the motor rotate at a desired speed.

$$\omega_{\text{desired}} = 1000 \text{ RPM}$$

The central question is: what should the motor command be if we want the speed to remain near this value?

2. Components of a Control System

A general control system may be represented as

$$\boxed{r(t)} \longrightarrow \boxed{\text{Controller}} \longrightarrow \boxed{u(t)} \longrightarrow \boxed{\text{Plant}} \longrightarrow \boxed{y(t)}.$$

For the DC motor speed-control system:

$$\boxed{\text{Desired RPM}} \longrightarrow \boxed{\text{Controller}} \longrightarrow \boxed{\text{PWM}} \longrightarrow \boxed{\text{DC Motor}} \longrightarrow \boxed{\text{Measured RPM}}.$$

Symbol	Meaning	Motor example
$r(t)$	Reference	Desired RPM
$u(t)$	Control input	PWM command or voltage command
$y(t)$	Output	Actual motor speed
$e(t)$	Error	Desired RPM minus actual RPM
Plant	System being controlled	DC motor and load
Controller	Rule or algorithm	Arduino or MATLAB control law

The plant is the physical system we want to control. In this course, the plant is the DC motor system.

3. Open-Loop Control

An open-loop system applies an input but does not use the measured output to correct that input.

$$\boxed{u(t)} \longrightarrow \boxed{G} \longrightarrow \boxed{y(t)}$$

For the motor:

$$\boxed{\text{PWM}} \longrightarrow \boxed{\text{Motor}} \longrightarrow \boxed{\text{RPM}}.$$

Suppose the following data are measured during one experiment:

PWM command	**Motor speed (RPM)**
50	300
100	650
150	1020
200	1350
255	1600

If PWM = 150 gives 1020 RPM, can we simply use PWM = 150 whenever we want 1020 RPM?
Not always. The motor speed can change even if the PWM command stays fixed.

4. Disturbances

A disturbance is an effect that changes the output but is not the command we intentionally apply.
For the motor, disturbances may include:

- load on the motor shaft;
- supply-voltage variation;
- friction;
- temperature;
- mechanical misalignment;
- differences between motors.

Without a disturbance:

$$\boxed{\text{150 PWM}} \longrightarrow \boxed{\text{Motor}} \longrightarrow 1020 \text{ RPM}.$$

With a load disturbance:

$$\boxed{\text{150 PWM}} \longrightarrow \boxed{\text{Motor + Load}} \longrightarrow 850 \text{ RPM}.$$

Limitation of open loop: If the output changes because of a disturbance, the command does not automatically change to correct the error.

5. Closed-Loop Feedback

A closed-loop system measures the output and uses that measurement to help determine the next input.

$$r(t) \longrightarrow \boxed{\sum} \longrightarrow e(t) \longrightarrow \boxed{C} \longrightarrow u(t) \longrightarrow \boxed{G} \longrightarrow y(t)$$

The output $y(t)$ is measured and fed back to the summing junction.
The error is

$$\boxed{e(t) = r(t) - y(t)}.$$

For motor speed control:

- $r(t)$ is desired RPM;
- $y(t)$ is measured RPM;
- $e(t)$ is speed error;
- the controller uses $e(t)$ to adjust PWM.

6. Worked Example: Motor Too Slow

Suppose the desired motor speed is

$$r = 1200 \text{ RPM}.$$

The measured speed is

$$y = 900 \text{ RPM}.$$

Then the error is

$$e = r - y = 1200 - 900 = 300 \text{ RPM}.$$

The error is positive, so the motor is too slow. A reasonable controller should increase the motor command.

A moment later, the measured speed becomes

$$y = 1150 \text{ RPM}.$$

Then

$$e = 1200 - 1150 = 50 \text{ RPM}.$$

The motor is still too slow, but the correction should now be smaller.

7. Worked Example: Motor Too Fast

Suppose the desired speed is still

$$r = 1200 \text{ RPM},$$

but the measured speed is

$$y = 1300 \text{ RPM}.$$

Then

$$e = 1200 - 1300 = -100 \text{ RPM}.$$

The error is negative, so the motor is too fast. A reasonable controller should reduce the motor command.

8. Disturbance Rejection

Disturbance rejection is one of the most important reasons to use feedback.

Suppose a motor is running near

$$1000 \text{ RPM}.$$

A load is suddenly applied.

Open-loop behavior

The PWM command remains fixed. The speed may drop:

$$1000 \rightarrow 800 \text{ RPM}.$$

Since the command does not change automatically, the motor may remain near 800 RPM.

Closed-loop behavior

The speed drops:

$$1000 \rightarrow 800 \text{ RPM}.$$

The error becomes

$$e = 1000 - 800 = 200 \text{ RPM}.$$

The controller increases the command, and the speed moves back toward 1000 RPM.

> Feedback does not prevent disturbances from occurring, but it gives the system a way to respond to them.

9. Basic Performance Vocabulary

Controls engineers need words to describe how a system behaves.

Term	Meaning
Reference	Desired value $r(t)$
Response	Actual output $y(t)$
Transient response	Behavior while the output is changing
Steady-state value	Value the output approaches after enough time
Steady-state error	Difference between reference and final output
Disturbance	External effect that changes the output

If

$$r = 1000 \text{ RPM}, \qquad y_{ss} = 950 \text{ RPM},$$

then the steady-state error is

$$e_{ss} = r - y_{ss} = 1000 - 950 = 50 \text{ RPM}.$$

10. In-Class Engineering Problem

A motor is commanded to operate at

$$r = 1500 \text{ RPM}.$$

The measured speed changes as follows:

$$y_1 = 1100 \text{ RPM}, \qquad y_2 = 1450 \text{ RPM}, \qquad y_3 = 1200 \text{ RPM}.$$

1. Calculate $e_1 = 1500 - y_1$.
2. Calculate $e_2 = 1500 - y_2$.
3. Calculate $e_3 = 1500 - y_3$.
4. At which point should the controller apply the largest corrective action?
5. Explain your answer in words.

Summary

- A control system makes a physical system behave in a desired way.
- Open-loop control does not use measured output to correct the input.
- Closed-loop control uses feedback.
- The feedback error is $e(t) = r(t) - y(t)$.
- Feedback helps the system respond to disturbances.
- DC motor speed control will be the main example throughout the course.